710001 POWER UNIT - in panel B2 (factory wired and tested; earthed aluminium enclosure)

INLET (outside B2)
powerCON TRUE1 TOP
UL 498, 20 A, locking
v1 position, X 120 Z 313

L N PE

W01
3 x 14
AWG

BREAKER 1P 15 A
UL 489, C curve;
protects the 14 AWG
mains wiring

GFCI RECEPTACLE
TR duplex NEMA 5-15R,
UL 943 Class A; unit
face, Z 1050 (D-70)

W02 L N

PE TERMINAL
inlet PE, receptacle
ground, PS1 FG, earth
stud, B2 frame bond W03

FG

W03 14 AWG green to B2's frame
(only B2 carries a mains part)

PS1 96 W CLG-100-24
kit feed, 24 V 4 A, Class 2
input 90-295 V, PFC
FG to the earth stud

F1 5 A
+24 V
0 V PS1

W02 L N

LAN
etherCON NE8FDP in B2,
W04 Cat6 to keystone 1,
keystone 2: W31 to the kit

PS2 60 W LPV-60-24
24 V 2.5 A, Class 2; light
controller, DMX512 in
Class II (no PE)

PS2 out
F2 5 A
+24 V, UP- DOWN- CALL-

RS-485

W02 L N

PS3 60 W LPV-60-12
12 V 5 A, Class 2
input 100-240 V
Class II (no PE)

+12 V, 0 V PS3

port 2

JOINTS AND HARNESS SEGMENTS (plug pair at every joint; cables clip to the cassettes only, D-26)

X1 unit / B2-CASS
X2 corner, in R1-CASS

W30b 4x16 AWG, 2 per pole B2-CASS
W30c R1-CASS

W10b B2-CASS
X4
W10c B4-CASS
X5

W20b air feed 4x18 AWG (2 per pole)
W20c R1-CASS
W20d

W20 and W24 share one 10-way plug at X1, X2 and X3

W24b hub link 6x24 AWG: RS-485, light current, rails
W24c
W24d

W31b Cat6 (RJ45 couplers)
W31c to the kit drop X7

X7 VOICE KIT DROP (R1, behind the screen)
one 24 V Class 2 feed, 5 A fuse: Mini-Fit Jr 4-way, capped
Cat6 keystone, capped
reached through felt plug FA-KIT

W11 in the shelf (6 x 18 AWG feed)

LIGHT SHELF (back and side wall tops, D-70)
uplight 4.40 m and down-light 2 x 0.95 m (24 V strips),
call drop at Y -209.5 (X6) down R2-CASS to the bar;
hub keeps up + down + call at about 22 W

X3 into the air column

AIR COLUMN (front-right inside corner; hub zone 180 x 180 at Z 1200, D-61, D-70)

FUSE BLOCK
behind the hub zone cover
F3a 2 A supply fans
F3b 2 A exhaust fans
F3c 1 A hub

DESK PRO AI SENSOR HUB
in-house, cited by name; CO2, presence
out: PWM 1, 2; RS-485 to the lights
in: tach 1, 2; inlet sensor; buttons;
light current; PS1-PS3 rails (H-ELE-05)

SUPPLY FANS S1 S2
high in DR; W40 4 x 20 AWG

EXHAUST FANS E1 E2
soffit right end; W41

INLET TEMP SENSOR
supply fan box inlet; W42

CONTROLS (hub zone)
BOOST, TEST, CALL buttons
LIGHT knob: all hub inputs
(W44); TEST = self-test D-87

CALL LIGHT (R2, above the screen)
900 mm bar, 2.7 W, on light controller CH3;
fed down from the shelf (X6, W12)
see the light shelf box

SAFETY AND RESERVES
SELF-TEST (D-87): TEST reads the 3 rails' volts, runs both fan pairs (tach), steps each light channel (current), reads the sensors and shows PASS or the failing item. PWM fail-safe: an open PWM line runs the fans at full speed.
Draw wire W33: back wall top -> ceiling cassette -> the mic and status light zone (D-41; also the sprinkler or alarm wiring point, ISS-052). Nothing is cut on site later (REQ-VCK-03).

NOTES
UNITS: mm. GENERAL TOLERANCE: harness lengths +20/-0 mm per segment; route positions +/-5 mm.
1. Mains (red) exists only inside B2 and the power unit (D-10, IF-12). Nothing mains is connected on site.
2. Every 24 V and 12 V circuit (blue) is Class 2: each supply is a Class 2 power unit on its own sheet (UL 1310), and no two meet (D-168).
3. Plug pairs: supply side receptacle (female), load side plug (male); Molex Mini-Fit Jr, keyed by way count.
4. Every cable through sheet steel runs in a grommet; cables clip to the cassette tray only (D-26).
5. Wire sizes and lengths: harness list (sheet 2 and the _SPEC.csv). Voltage drop under 3 %: calc-results.md.
6. Bought parts are named from their datasheets in the parts list 710000 (some still HOLD, each with its reason); no order, no supplier contact. The in-house hub is cited by name only.
7. Mains: US 120 V 60 Hz, 15 A, UL-listed parts (D-133); the prototype is tested in Vietnam through a 220-to-110 V step-down transformer.
8. Joints: X1 unit/B2-CASS X2 corner (R1-CASS) X3 column (P8, behind it) X4 B2/B4 X5 shelf X6 call X7 kit.

PROJECT	FocusPod v2 - Spaceship
TITLE	**ELECTRICAL - WIRING DIAGRAM**
DWG / PART NO.	710000 ELECTRICAL
REV / STATUS	P01 / DRAFT - not for production
UNITS / SCALE	mm / not to scale (schematic)
SHEET	1 of 3 (A1)
DATE / BY	2026-11-01 / Autonomous
SOURCE	electrical_model.py, electrical_parts.py

HARNESS LIST (every field, with conductors, routes, bushings and voltage drop: 710000_P01_ELECTRICAL-HARNESS-LIST_SPEC.csv)

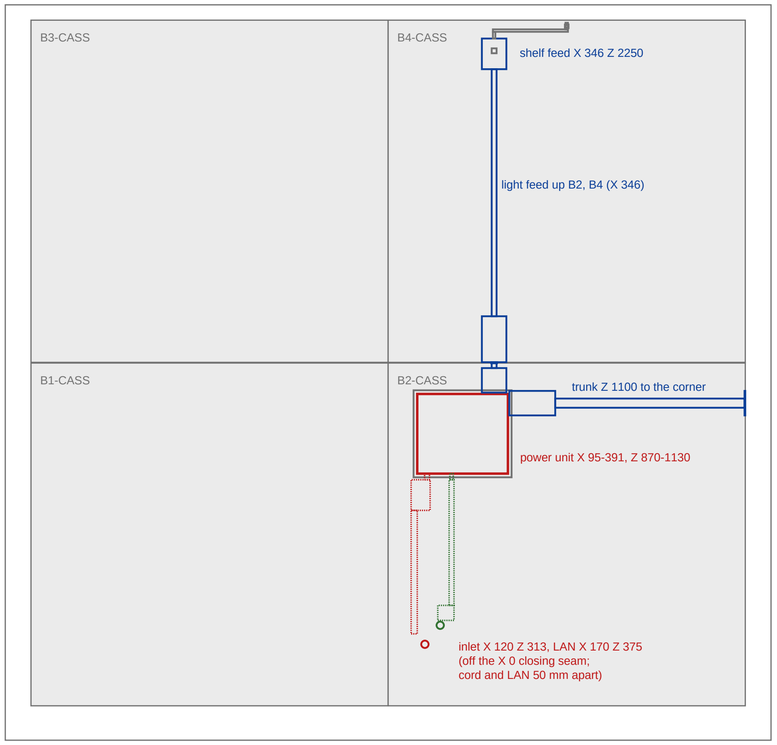
ID	SEGMENT	FUNCTION	CABLE	L mm	END A	END B
W01	PU	mains feed	Southwire SEOPRENE SJEOOW 14/3 cord, 300 V, 105 C (718089)	800	inlet terminals (powerCON TRUE1 TOP), 1/4 in disconnects (718074), inside the rear box 711003	breaker; the green core to the PE stud (H09)
W02	PU	mains inside the unit	UL 1015 14 AWG (black, white; green to the receptacle)	1500	breaker	GFCI receptacle LINE terminals; supply input leads in UL 486C lever connectors
W03	PU	protective bond	UL 1015 14 AWG green	300	M6 ring lug, 16-14 AWG	M6 ring lug, 16-14 AWG
W04	PU	LAN in	Cat6 U/UTP, Belden 3612 (23 AWG solid)	780	pressed into the etherCON's IDC (24-22 AWG solid); the jack built right-angled (Neutrik BDA 188), turned 180 degrees	punched down at keystone jack 1
W10a	PU	light feed	6 x UL 1015 18 AWG single cores, 105 °C (D-173)	170	light controller terminals	Mini-Fit Jr 6-way receptacle, just outside H03 (W10b's plug mates it in P5)
W10b	B2-CASS	light feed	6 x 18 AWG, Class 2	400	Mini-Fit Jr 6-way plug (mates W10a)	Mini-Fit Jr 6-way receptacle (tail out of the top grommet)
W10c	B4-CASS	light feed	6 x 18 AWG, Class 2	1360	Mini-Fit Jr 6-way plug (mates W10b in P4)	Mini-Fit Jr 6-way receptacle, tail into the shelf
W11a	SHELF	light shelf feed lead	6 x 18 AWG, Class 2	300	Mini-Fit Jr 6-way plug (mates W10c)	strip feeds: back uplight both ways; jumpers W11b, W11c
W11b	SHELF	jumper to the left wall uplight	2 x 18 AWG, Class 2	1610	strip feed connector	strip feed connector
W11c	SHELF	jumper to the right wall: uplight, down-light, call drop	4 x 18 AWG, Class 2	1220	strip feed connector	strip feed connectors; Mini-Fit Jr 2-way receptacle for the call drop (X6)
W20a	PU	air feed	4 x UL 1015 18 AWG single cores, 105 °C (D-173)	170	PS3 output	Mini-Fit Jr 10-way receptacle (with W24), just outside the unit's exit
W20b	B2-CASS	air feed	4 x 18 AWG (2 per pole), Class 2	1050	Mini-Fit Jr 10-way plug (with W24)	Mini-Fit Jr 10-way receptacle (with W24), tail through the corner slot S-CNR
W20c	R1-CASS	air feed	4 x 18 AWG (2 per pole), Class 2	1060	Mini-Fit Jr 10-way plug (with W24) (mated in P2 through the felt slot FS-CNR)	Mini-Fit Jr 10-way receptacle (with W24) in P8 behind the column, mated with W20d
W30a	PU	voice kit feed	4 x UL 1015 18 AWG single cores, 105 °C (D-173)	170	F1, 5 A (PS1, the kit's 24 V feed)	Mini-Fit Jr 4-way receptacle, just outside the unit's exit
W30b	B2-CASS	voice kit feed	4 x 16 AWG (2 per pole), Class 2	1050	Mini-Fit Jr 4-way plug	Mini-Fit Jr 4-way receptacle, tail through the corner slot S-CNR
W30c	R1-CASS	voice kit feed	4 x 16 AWG (2 per pole), Class 2	1050	Mini-Fit Jr 4-way plug (mated in P2 through the felt slot FS-CNR)	Mini-Fit Jr 4-way receptacle, capped with an empty plug housing (the kit outlet: one 24 V Class 2 feed, two pins per pole, 5 A fuse, 60 W peak)
W24a	PU	hub link to the power unit	6 x UL 1015 18 AWG single cores, 105 °C (D-173)	170	light controller link, PS2 current sense, PS1 and PS2 rail taps	shares W20's 10-way receptacle, just outside the unit's exit
W24b	B2-CASS	hub link to the power unit	6 x 24 AWG screened, Class 2	1050	shares W20's 10-way plug	shares W20's 10-way receptacle, tail through the corner slot S-CNR
W24c	R1-CASS	hub link to the power unit	6 x 24 AWG screened, Class 2	1060	shares W20's 10-way plug (mated in P2 through the felt slot FS-CNR)	shares W20's 10-way receptacle
W31a	PU	voice kit data	Cat6 U/UTP	300	punched down at keystone jack 2 (unit face)	RJ45 inline coupler
W31b	B2-CASS	voice kit data	Cat6 U/UTP	920	RJ45 inline coupler	RJ45 inline coupler, tail through the corner slot S-CNR
W31c	R1-CASS	voice kit data	Cat6 U/UTP	1050	RJ45 inline coupler (in P2)	RJ45 keystone jack, capped (the kit's data link)
W20d	COLUMN	air feed	4 x 18 AWG (2 per pole), Class 2	380	Mini-Fit Jr 10-way plug (with W24d): pulled out through FX-COL to mate W20c, pushed back into P8 before the column goes in	fuse block input (ferrules: they pass the 7.9 mm bushing)
W24d	COLUMN	hub link to the power unit	6 x 24 AWG screened	380	shares W20d's 10-way plug	hub I/O (HOLD H-ELE-05; ferrules, so it passes the 7.9 mm bushing)
W40	COLUMN	supply fan pair	4 x 20 AWG screened, plenum (CMP)	1910	fuse block and hub terminals	4-way PWM fan connector into a Y-cable at the fan pair (housing HOLD H-ELE-07)
W41	COLUMN	exhaust fan pair	4 x 20 AWG screened, plenum (CMP)	1550	fuse block and hub terminals	4-way PWM fan connector into a Y-cable (housing HOLD H-ELE-07)
W42	COLUMN	inlet temperature sensor	4 x 20 AWG screened, plenum (CMP)	2110	hub sensor connector (HOLD, ISS-060)	sensor probe (HOLD H-ELE-06)
W43	COLUMN	hub supply	2 x 18 AWG, Class 2	300	fuse block	hub power connector (HOLD, ISS-060)
W44	COLUMN	controls loom	6 x 24 AWG screened	300	button and knob tags	hub inputs (HOLD H-ELE-05)
W12	R2-CASS	call light drop	2 x 18 AWG, Class 2	1250	Mini-Fit Jr 2-way plug (mates W11c's call branch in P7)	strip feed of the call bar
W33	B4-CASS + CEILING	draw wire: ceiling route (D-41)	nylon draw cord 3 mm	1680	tied at P6	tied at the ceiling cassette's front hole

NOTES
UNITS: mm. GENERAL TOLERANCE: harness lengths +20/-0 mm per segment; route positions +/-5 mm.
1. Lengths: route model (calculation) + 150 mm loops and tails (assumption). The kit drop has a 500 mm service loop.
2. Plug pairs: supply side receptacle (female), load side plug (male). W20 and W24 share one 10-way at X1, X2, X3.
3. Every segment is tested in the factory before it ships (D-87): continuity, polarity, no shorts, latch pull.
4. Cables: Class 2 circuits on 300 V PVC cable (CL2 or CL3 listed for the US phase); inside the unit UL 1015 single cores, 105 °C (D-173).
5. W33 is a draw cord, tied off at each hole, for the voice kit's later cables to the mic zone (REQ-VCK-03).

PROJECT	FocusPod v2 - Spaceship
TITLE	**ELECTRICAL - HARNESS LIST**
DWG / PART NO.	710000 ELECTRICAL
REV / STATUS	P01 / DRAFT - not for production
UNITS / SCALE	mm / table, not to scale
SHEET	2 of 3 (A1)
DATE / BY	2026-11-01 / Autonomous
SOURCE	electrical_model.py, electrical_parts.py

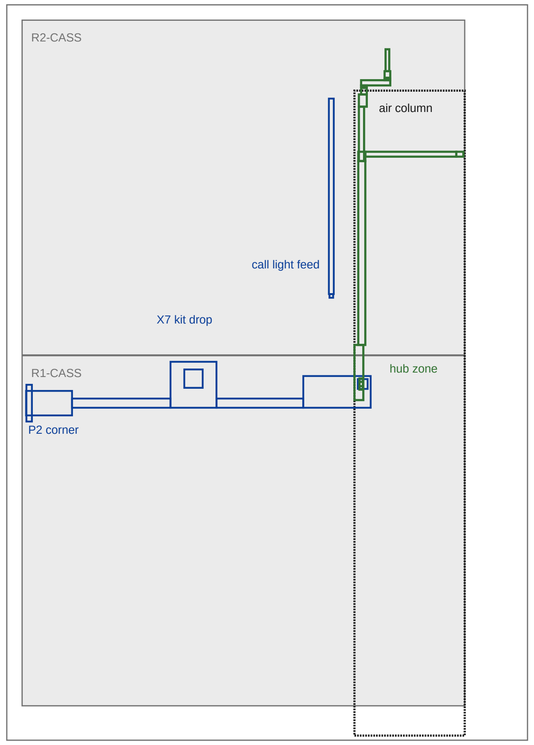
VIEW A - BACK WALL FROM INSIDE, looking toward +Y (1:10)

X view = +X, Y view = +Z; origin O on the room floor at the pod centre



VIEW B - RIGHT WALL FROM INSIDE, looking toward +X (1:10)

X view = -Y (front to the right), Y view = +Z



RESERVED SPACES IN THE CAD

Routes, pockets, bushings and slots are reserved spaces in the cassettes' 24 mm gap, in 710000_P01_ELECTRICAL_CAD, derived from skeleton P04. Pockets: P1 at the unit, P2 in R1-CASS behind the felt slot FS-CNR (hidden by B2-CASS's end), P4 above the B2/B4 joint, P5 and P6 the light feed, P3 the voice kit drop.

SITE ORDER (two people, one 5 mm key, no electrician)

1 Hang R1-CASS and R2-CASS; their tails stay out.
2 Hold B2-CASS near: plug its leads to the unit (X1) and to R1's corner tails (X2), push the pairs into P1 and P2, hang it.
3 B4-CASS: plug X4, push the pair into P4, hang it.
4 Plug the light shelf at X5 when the shelf goes up.
5 Before the air column goes in: pull X3 out of FX-COL, plug it, push it back into P8; plug X6 behind the column.
6 Plug the fans at their boxes: the supply pair through DR's opening (AIR's transition), the exhaust pair inside its box.
7 Room cord into the inlet, breaker on, press TEST: PASS (D-87).

KEEP-OUTS FOR OTHER TEAMS

WALL: B2 bay X 95-391, Z 870-1130 for the unit; the cord X 75-95 from Z 347, to X 137 from Z 750; the LAN X 162-215 Z 391-440, X 199-215 above; the holes and slots in B2-CASS, B4-CASS, R1-CASS, R2-CASS. INTERIOR: ledge and screen fixings 15 mm off the trunk band Z 1085-1115. AIR (CEILING): hub zone X 971-1151, Z 1110-1290; FR angle entries Z 1150, 1175; fill channel to Z 1890; tie pairs; 2 cordgrips; the ceiling cassette's holes at X 583 (D-41 route). FRONT: no hole in DR: the supply leads pass DR's opening.

ISO first angle

NOTES
UNITS: mm. GENERAL TOLERANCE: harness lengths +20/-0 mm per segment; route positions +/-5 mm.
1. Every coordinate in these views is computed from electrical_model.py (skeleton P04 parameters), so they follow the CAD.
2. Bushings: 2836N in the 19.1 holes (the unit's H01-H03, G-B2-B4 x2, G-CEIL-B/F), SB 437-5 in the FR angle (G-COL1, G-COL2) and the LAN's passage (the unit's H10, B2's lower member), SB 625-7 in the mains cord's (H04, the member); cordgrips at G-SOFFIT, G-FBOX; S-CNR gets an edge strip.
3. Felt-only holes (FX-*, FS-CNR, FA-KIT) go through the felt only, clear of the 12 mm lips: no steel edge, no bushing (no hole through a lip, D-69).
4. These views are the route drawing of 710000; there is no separate Fusion drawing of the route model (PM ruling 2026-11-01).

PROJECT	FocusPod v2 - Spaceship
TITLE	**ELECTRICAL - ROUTES AND JOINTS**
DWG / PART NO.	710000 ELECTRICAL
REV / STATUS	P01 / DRAFT - not for production
UNITS / SCALE	mm / 1:10 (views)
SHEET	3 of 3 (A1)
DATE / BY	2026-11-01 / Autonomous
SOURCE	electrical_model.py, electrical_parts.py